

LoRa Mesh for Large-Scale Networks: A Systematic Review of Routing, Energy, Coverage Distance, and Open Research Challenges

Maryam Ayad Kadhem
Department of Electrical Engineering
College of Engineering, University of Babylon
Hillah, Iraq
eng631.maryem.ayaad@student.uobabylon.edu.iq

Hilal Al-Libawy
Department of Electrical Engineering
College of Engineering, University of Babylon
Hillah, Iraq
Eng.hilal_al-libawy@uobabylon.edu.iq

Abstract—The existing LoRaWAN networks face problems because they can only operate in short distances, and they use one single transmission path which prevents their use in areas with remote locations or physical obstacles. This paper offers a complete assessment of how LoRaWAN technology developed from its initial single-hop system to its current multi-hop and mesh network structure. The study evaluates the routing performance and energy usage, modulation settings, and network expansion capabilities of LoRaWAN and LoRa Mesh systems. The study looks at how Spreading Factor (SF) and Chirp Spread Spectrum (CSS) modulation settings impact the distance transmission and data transmission capacity of LoRa devices. The review examines recent research on LoRa mesh networks that were published between 2022 and 2026, focusing specifically on the comparative findings that other studies reported. The surveyed systems utilize multi-hop communication as their primary method to increase coverage across different areas which includes 3 to 10 hops and beyond 10 hops. The reviewed studies demonstrate that adaptive spreading factor selection, commonly between SF7 and SF12, plays a key role in balancing transmission range and energy consumption. The majority of research studies utilize either energy-aware routing or lightweight routing systems, which enable them to communicate over distances of a few kilometers up to more than ten kilometers while using minimal power. Through a detailed review of existing routing protocols including reactive, proactive, and hybrid approaches. From the analysis of previous studies, we conclude that the spreading factor is directly proportional to the communication distance; specifically, the transmission range extends with each increasing spreading factor, while the higher spreading factor values enhance the ability to detect signals in very noisy conditions.

Keywords—CSS modulation, energy efficiency, IoT, LoRaWAN, LoRa mesh, LPWAN, multi-hop routing, wireless networks

I. INTRODUCTION

This capability makes LoRaWAN particularly appealing for applications like smart agriculture, where sensors can be deployed over vast areas to monitor soil conditions and crop health. As cities continue to grow, the demand for reliable long-range communication solutions will likely increase, paving the way for further advancements in this technology. The general wireless systems which serve this purpose include Wi-Fi, ZigBee, and Bluetooth, which provide operational ranges between several meters and tens of meters. The systems enable organizations to extend their coverage through the use of mesh network topologies which support multi-hop communication [1].

By considering the unique constraints of subterranean environments, such as signal attenuation and energy

limitations, this strategy optimizes route selection for better performance in these complex networks. Landing effects are less affecting. For this kind of system (LPWANs), sensor modules are the at-issue items. have been emerging as a promising alternative to extend coverage in these challenging environments [2].

One of the most widely adopted LPWANs is LoRaWAN [3], providing a low-cost way to ensure the data reaches long ranges [3].

Mesh LoRa networks use IoT devices to send messages between different network nodes. The multi-hop transport system of mesh networks enables transmission data to travel over a wider area between any two network nodes. Mesh LoRa networks offer network services that enable technology to run new application types that operate on IoT nodes [4].

This evolution towards nested LoRa networks enables the transmission and reception of LoRa data as relays, creating large-scale nested networks for transmission in multi-obstacle environments, such as indoor forested smart city networks [5].

Mesh Connect AP Nodes provide a Wi-Fi LoRa mesh network. Mesh Connect enables end-users in remote locations to connect to the Internet via Wi-Fi without the need for additional hardware or software and with minimal infrastructure investments [6]. In rural and subterranean environments, deploying and maintaining such LoRaWAN networks is often impractical due to limited telecommunication infrastructure and challenging terrain. As a result, multi-hop and mesh LoRa networks, enabled by LoRa repeaters (RPs), have been emerging as a promising alternative to extend coverage in these challenging environments [2].

the second section of this document presents an overview of wireless network systems and LoRa technology basic principles. The third section of the document provides the theoretical background which includes SF and CSS modulation. The study analyzes routing protocols used in LoRa mesh networks through its fourth section. The fifth section of the document presents a comparative study of recent research work. The sixth section of the document ends the paper while showing potential research paths for upcoming studies.

II. WIRELESS NETWORK SYSTEM

Each of the wireless communication technologies , offers advantages in conditions of data rate, coverage, and power consumption. The three types of distances can be categorized as short-range, long-range, and medium-range as shown in Table I. Short-range solutions, like Wi-Fi, Bluetooth, Zigbee,

and UWB, operate from a few meters to tens of meters, ideal for wearables, smart homes, and industrial automation [7].

TABLE I. PERFORMANCE COMPARISON OF SHORT-RANGE WIRELESS TECHNOLOGIES

Standard Communication	Wi-Fi	Bluetooth	ZigBee	UWB
IEEE defined Standard	802.11a/b/g	802.16.1	802.15.4	802.15.3a*
Typical range	100 m	10 m	10-100 m	10 m
Operating frequency band	2.4 GHz; 5 GHz band	2.4 GHz band	868/915 MHz; 2.4 GHz band	3.1-10.6GHz band

III. LoRa (LONG RANGE)

LoRaWAN technology serves as a primary component for Internet of Things (IoT) systems because it provides two-way communication and location tracking functions. The system operates using low power consumption while delivering extended distance communication capabilities. the system operates as a low-power wide-area network which enables remote communication through its design for extended distances while using minimal power and transmitting data at slow speeds [8].

It utilizes modulation, wide-bandwidth (BW), base-reading factor (SF), frequency band, coding learning (CR), and other capabilities, which is why it is used to configure LoRa radio modules [5, 6]. A chirp spread spectrum (CSS) modulation method is used, providing better noise immunity and an increased channel budget.

A LoRa mesh network is a multi-dependent network based on LoRa technology as shown in Table II. The core network relies on the LoRaWAN protocol with some modifications and additions that operate on the packet protocol, wireless network, and a network server to retransmit packets over multiple hops. Fig. 1 provides an overview of the networks.

The medical (ISM), scientific, and industrial frequency band that LoRa devices operate on is 868 MHz in Europe and 915 MHz in the US [10].

TABLE II. COMPARISON BETWEEN LoRaWAN AND LoRa MESH

LoRa WAN	LoRa Mesh
Single-hop star topology (MAC)	Multi-hop routing using Relay/Router nodes to extend reach
Strict duty cycle limits	Predictive wake-up, time slotting, or adaptive channel sensing (LBT).
ALOHA-based MAC (No CSMA)	Collision avoidance mechanisms, concurrent transmission exploitation (SF orthogonality)
Low bandwidth (0.25–11 kbps)	Minimal control packet size/frequency; reliance on lightweight distance-vector metrics
It contains only a single central gateway	It not contains a central gateway but contain many nodes and multi-hop communication.

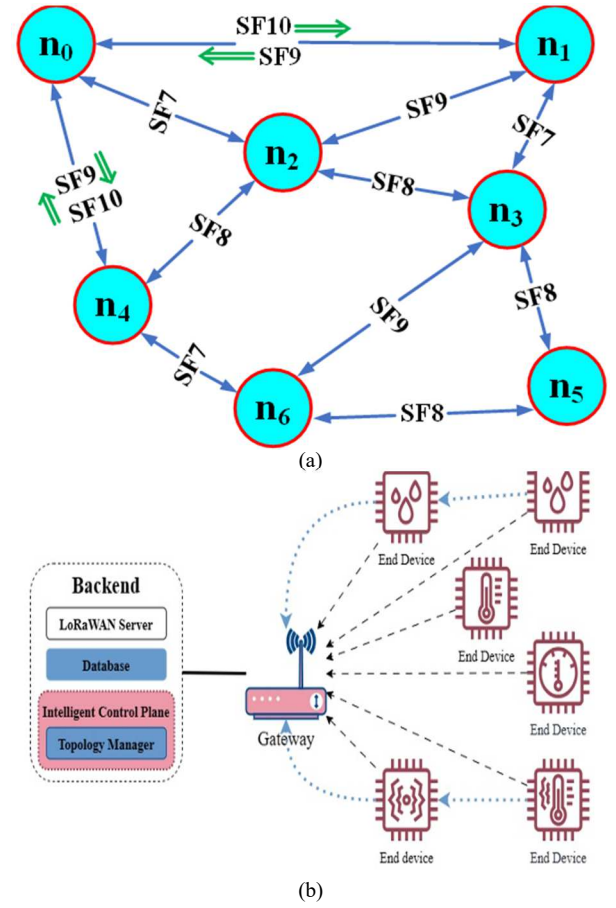


Fig. 1. LoRa-based mesh network structure vs. structure of LoRaWAN network; (a) LoRa mesh network structure (b) architecture of a LoRaWAN network [9].

The diagram shows how LoRa mesh networks differ from conventional LoRaWAN designs. According to the LoRa mesh topology (see Fig. 2a), network nodes function as both end devices and relay points to establish multi-hop network connections. The system enables data packet transmission through various intermediate nodes which allows for extended network reach while maintaining better connectivity in remote locations and areas with physical barriers.

The LoRaWAN system operates using a star topology through which end devices send data directly to a central gateway as shown in Fig. 2b. The single-hop communication system provides easier network setup but its coverage and scalability are restricted in situations where users cannot directly communicate with the gateway.

IV. THEORETICAL BACKGROUND

LoRa technology is a relatively new and emerging technology currently available on the market, operating in the unlicensed sub-1 GHz band for long-range communication. LoRa is the company's proprietary Dispersive Spectrum Modulation (DSM) technology, derived from CSS, that balances data throughput and sensitivity within a fixed channel frequency range. It was generally used in military applications because of its long distance in communication and ability to prevent interference. implementation for commercial usage [9].

A. SF (Spreading Factor) and CSS (Chirp Spread Spectrum)

This parameter is considered one of the most important parameters in LoRa technology. This is the spreading factor (SF) since it uses CSS modulation to determine the spectral density. When SF is higher, it means better reception sensitivity and longer transmission ranges; another daily advantage of using Comcast on the high transmitting ends would be an increase in their counterparty's own SF. Fig. 3. In the previous scenarios, the user contract was configured to use 7 to 12 SFs randomly (most popularly being usually used in the Lora racket). rate application. These are arguments for the highly stress-meaning parameters) [10].

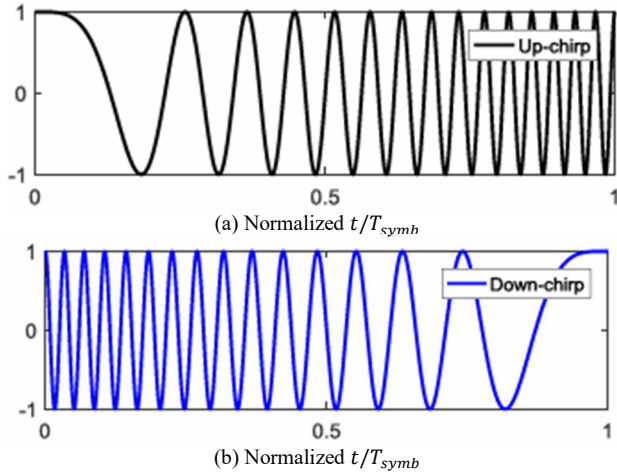


Fig. 2. Upward and downward chirp signals normalized to the symbol time (T_{symb}) [11].

$$P(SFs) = \frac{2SF_{max} - SF_n}{\sum 2SF_i - SF_{min}} \quad (1)$$

Since:

$$\sum P(SF_i) = 1$$

$$P(SF_{min}) = 2SF_{max} - SF_{min} \cdot P(SF_{max})$$

Where SF_{max} is the maximum spreading factor and SF_{min} is the minimum spreading factor [12].

CSS (Chirp Spread Spectrum) uses an embedding method that defends against automated trail fading attacks and operates effectively in noisy settings. The system enables outdoor connections that exceed 10 kilometers in rural areas to maintain low power usage while delivering essential low-voltage links. Its characteristics were analyzed in detail [13].

Chirp Spread Spectrum (CSS) modulation that is LoRa technology-based on a patent by SimTech and that enables data to transmit up to several kilometers. Chirp Spread Spectrum (CSS) modulation uses chirp signals which Fig. 3 demonstrates through its display of upward and downward chirps that researchers have normalized to show their patterns throughout each symbol duration. An upward chirp conducts a signal frequency ascent which continues until the end, while a downward chirp sees frequency decline throughout its duration. The CSS modulation system depends on this frequency variation which serves as its primary characteristic to maintain reliable connections in environments with high

levels of noise and interference. the transmission range achieved when communicating using LoRa devices mainly depends on SF and TX power. The SF, whose spreading factor values are between 7 and 12, It boasts reliable protection with a higher diffusion coefficient and higher sensitivity and can return with fewer errors but requires very high transmission effort: this is a range-limiting connection at the expense of a lower data rate and higher power consumption. Fig. 3 show how spreading factor selection and chirp characteristics determine system performance [13].

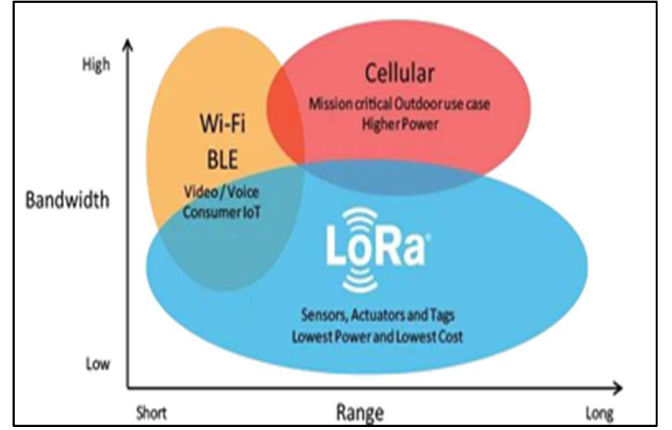


Fig. 3. CSS in LoRa compared with other wireless technology [14].

In this modulation technique, The pulses are sinusoidal signals. whose frequencies increase or decrease include a defined range and at a specific rate. The modulation parameters are [15]:

- Transmission power Frequency,
- Bandwidth,
- Coding rate,
- Spreading Factor (SF),
- Frequency.

The modulation equation can be represented as [16]:

$$S(t) = \cos[2\pi(f_0 + kt)t] \quad (2)$$

Where f_0 is the first value of frequency and k is the rate of change of frequency.

B. Routing Protocol

Studies have indicated that multi-faceted routing improves and reduces power consumption. Some routing protocols are being developed from other standards, such as Bluetooth, enabling LoRa networks to transcend their original architectural design [17].

The routing system handles the task of sending data packets through different pathways in the network. As shown in Table III, routing establishes network connections through the evaluation process, which assesses all possible routes from a source node to find the best route.

The criteria for determining the best paths are battery charge level, number of hops, link status, and geographical position [18].

TABLE III. COMPARATIVE ANALYSIS OF KEY LoRa MESH ROUTING PROTOCOLS

Protocol Name	Classification	Key optimization mechanism	Primary Optimization Focus	Reference
Lora Blink	Synchronous flooding	The main protocol integrates MAC and routing in asynchronous flooding mechanism.	Enhanced coverage achieves a reliable packet delivery rate	[1]
LoRa mesh	Distance factor	The Ans-3 model enables users to create simulations which they can use to assess performance differences between multi-hop LoRa networks and single-hop LoRaWAN networks.	PDR extension at network edge.	[22]
In Dynamic Source Routing (DSR) and Distance Vector Routing (DVR), the preferred mode is Ad hoc on-demand distance vector routing	Reactive and Active	Comparison between DVR, AODV, and DSR	These charges are ineffective for handling high node mobility.	[20]
DSR	Reactive	Designed for use in a dedicated multi-gateway wireless network.	The network is allowed to be self-configured and self-organized, without the need for any other existing infrastructure or management.	[16]
AODV and Hybrid Wireless Mesh Protocol (HWMP)	Hybrid	Transmission efficiency and node lifetime.	AODV and Hybrid Wireless Network Protocol (HWMP)	[10]

The modified protocols for LoRa networks can be divided into three types [19]:

- **Proactive protocols:** These protocols keep updated routes in constant route tables. Though they allow for instant routes, they require excessive bandwidth, which makes them inappropriate for LoRa's resource constraints.
- **Reactive Protocols:** Routes are known only when a packet of data needs to be transmitted. This is most beneficial in terms of scalability, as less power is used in reactive protocols, as routes tend to update less often.
- **Hybrid Protocols:** An attempt to blend proactive route maintenance for local routes and reactive route discovery for far routes for an efficient mix of both approaches. However, research reveals that hybrid protocols benefit only in very scenario-specific topological layouts, failing to provide efficiency for large, dynamic, large-scale IoT deployments that behave unpredictably.

V. LoRa MESH RELATED WORK

Recent studies between 2022 and 2025 have focused on enhancing the limitations of traditional single-hop LoRaWAN systems by introducing multi-hop and mesh-based communication frameworks as shown in Table IV. Trajano et al. in 2022 [6], proposed the integration of Wi-Fi over LoRa mesh to provide network access in remote and infrastructure-poor regions, with moderate energy consumption. In 2023, Ong and Phan [14], explored a reactive ad-hoc routing mechanism that improved route discovery efficiency while maintaining low power use.

Similarly, Tian et al. in 2023 [13], introduced LoRaHop, a protocol extending LoRaWAN to support multi-hop communication for both uplink and downlink messages,

achieving greater coverage without the need for additional gateways.

Hammouti et al. in 2024 [5], conducted a study which showed that open-source LoRa mesh technologies provide superior scalability and energy savings compared to proprietary LoRa mesh technologies. Hakani et al. in 2024 [7], built upon this concept by demonstrating that LoRa mesh technology functions as a reliable solution for remote areas which lack Internet access. The researchers, Pueyo Centelles et al. in 2024 [12], created an energy-efficient distance-vector routing system which enables flexible network operation across different types of terrain.

The research of Durand and Booyesen in 2025 [20], evaluated mesh-topology performance while demonstrating that adaptive duty cycles deliver energy savings for large-scale IoT systems. Udugampola et al. in 2025 [2], and Miquel Solé et al. in 2025 [4], presented their latest research, which focuses on two specific routing methods: energy-aware routing and position-based routing to improve system scalability.

The research shows that cross-layer adaptive routing systems have become the preferred approach because they achieve energy efficiency and enable longer distances and autonomous functioning in complex IoT setups.

The research demonstrates a major shift from traditional LoRaWAN systems toward advanced self-organizing mesh networks which can function in actual field situations.

From the analysis of previous studies, we conclude that the spreading factor is directly proportional to the communication distance; specifically, an increase in the spreading factor leads to a corresponding increase in transmission range. Higher spreading factor values significantly enhance signal detectability in severe noise environment's.

TABLE IV. COMPARATIVE ANALYSIS OF PREVIOUS STUDIES (2022-2026)

Reference	Author	Year	Protocol/Technology	Range/Distance	SF (Spreading Factor)	Power Characteristics	No. of Nodes/Hops
[6]	Trajano et al.	2022	Wi-Fi over LoRa Mesh (Mesh Connect AP Nodes))	1-2 Km (urban remote zones)	SF 7-9 typical	Moderate power consumption optimized for gateway nodes	Multi-hop structure (3-5 hops)
[14]	Ong & Phan	2023	Reactive Ad-hoc routing (LoRa mesh node design)	Up to 3 km	SF 9–11	Adaptive power per route discovery; energy-efficient under low mobility	Multi-hop routing; 5–8 nodes
[13]	Tian et al.	2023	LoRaHop – Multi-hop LoRaWAN (uplink & downlink)	Several Km (rural/suburban)	SF 8–12 adaptive	Transmission power dynamically adjusted (TX 14–20 dBm)	3-hop maximum
[5]	Hammouti.et al.	2024	Open source & proprietary LoRa mesh	5–7 km (open field)	SF 7–10	Energy-aware routing; reduced consumption up to 40%	Multi-hop, >10 nodes
[7]	Hakani et al.	2024	LoRa mesh for internet-deprived areas	8–10 km (mountainous terrain)	SF 10–12	Optimized low-power operation for isolated nodes	Multi-hop, 6–12 nodes
[12]	Pueyo Centelles et al.	2024	Minimalistic distance-vector routing (LoRa mesh)	4–6 km	SF 7–11	Low-energy design using lightweight vector metrics	Multi-hop adaptive topology
[20]	Durand & Booyesen	2025	Mesh-topology LoRa network (performance evaluation)	6–8 km (semi-urban)	SF 8–10	Energy-aware scheduling with adaptive duty cycles	3–5 hops
[2]	Udugampola et al.	2025	PEAR – position & energy aware routing	10–12 km (subterranean / rural)	SF 9–12 adaptive	Power optimized; up to 75% reduction vs. conventional	Multi-hop 10+ nodes
[4]	Miquel Solé et al.	2025	Middleware for distributed LoRa mesh applications	5–7 km	SF 7–10	Dynamic power scaling based on node load	Multi-hop with distributed nodes
[21]	Bhat et al.	2026	LoRaDisC (LoRa Distributed Scheduling and Coordination)		SF 7-12	Battery-powered)	2-6 hops

VI. CONCLUSION

The review demonstrates that LoRa technology now supports multiple network architectures which use multi-hop and mesh networks instead of its initial single-hop communication design. The evolution is driven by real deployment challenges which arise when LoRaWAN systems face difficulties in rural areas and underground spaces and areas with dense obstacles. LoRa mesh networks enable nodes to function as relay points which boosts their network range and connectivity capacity without needing full support from centralized gateways.

The studies which researchers examined show that LoRa mesh systems depend on two main factors for their network performance which include selected routing methods and basic physical layer characteristics with special focus on spreading factor and transmission power. Energy-aware and location-based routing approaches appear to be especially effective in balancing communication range, and battery lifetime. The analysis of recent related work shows that LoRa mesh networks now function as a practical method to extend low-power wireless communication beyond the range of traditional single-hop LoRaWAN systems. The related work

table summarizes studies which demonstrate that multi-hop configurations with multiple nodes and adaptive routing can extend communication range while using less energy. The reported deployments can maintain reliable connections over distances which go from a few kilometers to more than ten kilometers depending on the selected spreading factor and terrain conditions.

The reviewed works demonstrate that researchers consistently depend on adaptive spreading factors together with energy-aware routing mechanisms to achieve optimal node lifetime and packet delivery performance. The findings demonstrate that lightweight and context-aware routing approaches. In conclusion, recent studies demonstrate that LoRa mesh networking provides businesses with a communication system which can grow and use energy-efficient and flexible solutions that meet their future needs for IoT deployments in large-scale and infrastructure-limited environments. LoRa mesh networking provides organizations with a valuable solution which enables them to implement large-scale IoT systems that require dependable operations and extensive coverage at minimal energy costs.

The ongoing development of routing systems and cross-layer optimization techniques for routing systems will transform networks into autonomous systems with scalable capabilities. I found through my study of previous research that the spreading factor determines how far a signal can travel because whenever the spreading factor increases the transmission range also increases. Higher spreading factor values significantly enhance signal detectability in severe noise environments.

REFERENCES

- [1] J. R. Cotrim and J. H. Kleinschmidt, "LoRaWAN mesh networks: A review and classification of multihop communication," *Sensors*, vol. 20, no. 15, p. 4273, 2020.
- [2] N. Udugampola, X. Ai, B. Li, H. Gong, and A. Seneviratne, "A Position-and Energy-Aware Routing Strategy for Subterranean LoRa Mesh Networks," in *2025 IEEE 50th Conference on Local Computer Networks (LCN)*, Sydney, Australia, 2025: IEEE, pp. 1-9.
- [3] N. C. Almeida, R. P. Rolle, E. P. Godoy, P. Ferrari, and E. Sisinni, "Proposal of a hybrid LoRa Mesh/LoRaWAN network," in *2020 IEEE International Workshop on Metrology for Industry 4.0 & IoT*, Roma, Italy, 2020: IEEE, pp. 702-707.
- [4] J. Miquel Solé, R. Pueyo Centelles, F. Freitag, R. Meseguer, and R. Baig, "Middleware for distributed applications in a lora mesh network," *ACM Transactions on Embedded Computing Systems*, vol. 24, no. 4, pp. 1-26, 2025.
- [5] M. Hammouti, O. Moussaoui, M. Hassine, and A. Betari, "Exploring open source and proprietary LoRa mesh technologies," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 34, no. 2, pp. 960-969, 2024.
- [6] A. F. Trajano, P. C. Andrade, I. A. Pimenta, J. Cavalcante, and J. N. de Souza, "Delivering WiFi connectivity to remote locations through LoRa mesh networking," in *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium*, 2022: IEEE, pp. 1-6.
- [7] R. Hakani, A. Mishra, N. Shah, and A. Rawat, "Enhancing situational awareness with lora mesh networks: communication in internet-deprived areas," 2024.
- [8] D. Lundell, A. Hedberg, C. Nyberg, and E. Fitzgerald, "A routing protocol for LoRa mesh networks," in *2018 IEEE 19th International Symposium on "A World of Wireless, Mobile and Multimedia Networks"(WoWMoM)*, Chania, Greece, 2018: IEEE, pp. 14-19.
- [9] R. S. Sinha, Y. Wei, and S.-H. Hwang, "A survey on LPWA technology: LoRa and NB-IoT," *ICT Express*, vol. 3, no. 1, pp. 14-21, 2017.
- [10] R. Pueyo Centelles, "Towards LoRa mesh networks for the IoT," Doctoral dissertation, Polytechnic University of Catalonia, 2021.
- [11] V. Fialho and F. Azevedo, "Wireless communication based on chirp signals for LoRa IoT devices," *i-ETC: ISEL Academic Journal of Electronics Telecommunications and Computers*, vol. 4, no. 1, p. 6, 2018.
- [12] R. P. Centelles, R. Meseguer, F. Freitag, R. B. Viñas, and L. Navarro, "A minimalistic distance-vector routing protocol for lora mesh networks," *IEEE Access*, vol. 12, pp. 128941-128962, 2024.
- [13] P. Tian, C. A. Boano, X. Ma, and J. Wei, "LoRaHop: Multihop support for LoRaWAN uplink and downlink messaging," *IEEE Internet of Things Journal*, vol. 10, no. 17, pp. 15376-15392, 2023.
- [14] M.-D. Ong and T.-M. Phan, "Mesh LoRa Node Design with Reactive Ad-Hoc Routing Protocol," *Journal of Aviation Science & Technology*, vol. 2, no. 1, pp. 10-15, 2023.
- [15] W. D. Paredes, H. Kaushal, I. Vakulinia, and Z. Prodanoff, "Lora technology in flying ad hoc networks: A survey of challenges and open issues," *Sensors*, vol. 23, no. 5, p. 2403, 2023.
- [16] L. Aldaheri *et al.*, "LoRa communication for agriculture 4.0: Opportunities, challenges, and future directions," *IEEE Internet of Things Journal*, vol. 12, no. 2, pp. 1380-1407, 2024.
- [17] A. W.-L. Wong, S. L. Goh, M. K. Hasan, and S. Fattah, "Multi-hop and mesh for LoRa networks: Recent advancements, issues, and recommended applications," *ACM Computing Surveys*, vol. 56, no. 6, pp. 1-43, 2024.
- [18] D. de Farias Medeiros, M. R. Villarim, F. B. S. de Carvalho, and C. P. de Souza, "Implementation and analysis of routing protocols for lora wireless mesh networks," in *2020 11th IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, BC, Canada, 2020: IEEE, pp. 0020-0025.
- [19] A. A. Al-Healy and Q. Ibrahim, "Evaluation Metrics and Optimization Strategies for Routing Protocols in Resource-Constrained Wireless Sensor Networks," 2025.
- [20] T. G. Durand and M. J. Booyesen, "Performance Evaluation of a Mesh-Topology LoRa Network," *Sensors*, vol. 25, no. 5, p. 1602, 2025.
- [21] S. A. Bhat, I. B. Sofi, M.-C. Chen, and N.-F. Huang, "RF-Driven Adaptive Surrogate Models for LoRaDisC Network Performance Prediction in Smart Agriculture and Field Sensing Environments," *AgriEngineering*, vol. 8, no. 1, p. 27, 2026.